

4(a)	(electric) force is (directly) proportional to product of charges	B1
	force (between point charges) is inversely proportional to the square of their separation	B1
4(b)(i)	arrows showing tension upwards in direction of string, electric force horizontally to the right and weight vertically downwards and all three labelled	B1
4(b)(ii)	$F_E = \frac{96 \times 10^{-9} \times 64 \times 10^{-9}}{4 \times \pi \times 8.85 \times 10^{-12} \times 0.080^2}$ $ (= 8.63 \times 10^{-3} \text{ N})$	C1
	either angle to vertical $= \sin^{-1} 0.080 / 1.2$ $ (= 3.82^\circ)$	C1
	weight $= F_E / \tan 3.82 = 8.63 \times 10^{-3} / \tan 3.82$ $ (= 0.129 \text{ N})$	C1
	mass $= 0.129 / 9.81$ $ = 0.013 \text{ kg}$	A1
	or $T \sin \theta = mg$ and $T \cos \theta = F_E$ or $\tan \theta = mg / F_E$	(C1)
	$\tan \theta = 1.2 / 0.080$	(C1)
	$m = (1.2 \times 8.63 \times 10^{-3}) / (0.080 \times 9.81)$ $ = 0.013 \text{ kg}$	(A1)

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Question	Answer	Marks
4(b)(iii)	$E_p = \frac{Q_1 Q_2}{4\pi\epsilon_0 r} = \frac{96 \times 10^{-9} \times 64 \times 10^{-9}}{4 \times \pi \times 8.85 \times 10^{-12} \times 0.080}$ $= 6.9 \times 10^{-4} \text{ J}$	A1
4(c)(i)	towards the top of the page / towards plate P	B1
4(c)(ii)	$F = QE$ and $E = V/d$	C1
	$F = 1.6 \times 10^{-19} \times 250 / 0.018$ $= 2.2 \times 10^{-15} \text{ N}$	A1
4(c)(iii)	either the force is not (always) perpendicular to the velocity or the force is always in the same direction	B1